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Occurrence of fungal contaminants and polychlorinated biphenyls in coastal groundwater of Ondo State, Nigeria: implications for water quality

Felix Adeleke Ikuesan¹ and Thompson Faraday Ediagbonya^{2,3*}

*Correspondence:

Thompson Faraday Ediagbonya
thompson.faraday@unilak.ac.rw

¹Department of Biological Sciences,
Olusegun Agagu University of
Science and Technology, Okitipupa,
Ondo State, Nigeria

²Department of Chemical Sciences,
Olusegun Agagu University of
Science and Technology, Okitipupa,
Ondo State, Nigeria

³Faculty of Environmental Studies,
University of Lay Adventists of
Kigali, Kigali, Eastern District,
Rwamagana, Rwanda

Abstract

Access to safe drinking water remains a challenge in many rural coastal communities. This study assessed fungal contaminants and polychlorinated biphenyls (PCBs) in groundwater from coastal Ondo State, Nigeria, and evaluated associated public health risks. Using culture-based methods and GC-MS analysis, twelve fungal isolates across eight genera, including *Aspergillus*, *Fusarium*, and *Penicillium*, known for mycotoxin production were detected. Among the fungi, *Rhodotorula* and *Rhizopus* had the highest percentage occurrence of 77.8 and 66.7% respectively. Eighteen PCB congeners were quantified, with individual congener concentrations ranging from below the detection limit to 3.90 ng/L (PCB18); the corresponding total (Σ PCB) concentration is reported in the main text. Low-chlorinated homologues (Di- and Tri-PCBs) dominated the congener profile. All PCB concentrations remained below the US EPA Maximum Contaminant Level of 500 ng/L for total PCBs. Corrected non-carcinogenic health risk assessments revealed that hazard quotients for all assessed congeners remained below the threshold of 1.0, indicating that modelled non-carcinogenic risks are within acceptable limits; carcinogenic risk was not evaluated because the required congeners were not determined. Nevertheless, the occurrence of fungal genera at colony counts exceeding commonly cited but non-regulatory reference values warrants further microbiological monitoring; fungal pathogenicity, toxin production and health outcomes were not directly assessed in this study. These findings highlight the need for integrated monitoring of microbial and chemical contaminants in coastal groundwater systems and further investigation to inform targeted interventions to protect community health.

Highlights

- First reported integrated assessment of fungal genera and PCBs in coastal groundwater of Ondo State.
- Twelve fungal isolates across eight genera detected including mycotoxin-producing *Aspergillus* and *Fusarium*.
- Eighteen PCB congeners identified with concentrations remaining below US EPA guidelines.



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- Low-chlorinated PCB homologues (Di- and Tri-PCBs) dominated the congener profile.
- Corrected risk assessments indicate non-carcinogenic hazard quotients are within acceptable limits.

Keywords Coastal communities, Fungi, PCBs, Risk assessment, Groundwater

1 Introduction

The consumption of microbiologically contaminated water remains a major global public health concern, contributing significantly to endemic diseases such as cholera, typhoid fever, diarrhea, tuberculosis and hepatitis A [11, 23]. While bacteria, viruses and parasites are widely recognized as important waterborne pathogens, fungi have received comparatively less attention despite their increasing detection in drinking water systems. Fungi can alter organoleptic properties of water and produce mycotoxins with documented health effects [14]. These organisms, which include yeasts and filamentous molds, are heterotrophic and capable of producing secondary metabolites with potential toxicological significance. Their presence in drinking water has been linked to contamination during pipe installation, distribution system biofilm formation and poor maintenance of water infrastructure. In immunocompromised individuals, exposure to waterborne fungi may contribute to opportunistic infections and other health complications [14].

Beyond microbial contamination, groundwater resources are increasingly threatened by chemical pollutants, particularly persistent organic pollutants (POPs). These compounds can be introduced into aquatic systems through industrial discharge, agricultural runoff and improper waste disposal practices. Among POPs, polychlorinated biphenyls (PCBs), dichlorodiphenyltrichloroethane (DDT) and dichlorodiphenyldichloroethylene (DDE) are of major concern due to their toxicity, environmental persistence and ability to bioaccumulate in biological systems [1, 27, 28]. While various POPs exist in the environment, this study focuses exclusively on PCBs due to their extreme persistence, widespread historical use in electrical equipment in Nigeria, and high potential for groundwater leaching compared to other POPs that may degrade faster or sorb more strongly to surface soils.

The potential sources of both microbiological and chemical contaminants in the study area are largely linked to anthropogenic activities. Microbiological contamination primarily originates from poorly constructed hand-dug wells, proximity to pit latrines, and surface runoff carrying animal waste. Conversely, chemical contaminants such as PCBs are potentially introduced through improper disposal of spent transformer oils, e-waste dumping, and leachate from unlined municipal landfills. Furthermore, the co-occurrence of these contaminants warrants investigation, as microbial communities can influence the biodegradation and transformation of persistent organic pollutants in groundwater systems. These shared anthropogenic pressures create a complex contamination profile in coastal aquifers.

Groundwater, which forms a critical component of the hydrological cycle within subsurface aquifers, serves as a major source of drinking water in many developing regions [33]. However, groundwater contamination is often difficult and costly to remediate, posing long-term risks to both environmental and human health [37, 38]. In coastal and peri-urban settings, proximity to waste dumps, mechanic workshops

and poorly managed landfills significantly increases the risk of groundwater pollution due to leachate infiltration [30, 31]. Recent studies in West Africa have reported deteriorating groundwater quality linked to both microbial and chemical contaminants. For instance, Ayeta et al. [6] observed elevated health risks associated with groundwater use in coastal Ghana, while Houédakor et al. [17] reported spatial variability in groundwater quality in coastal Togo. Similarly, Ikuesan and Ediagbonya [21] and Nlemolisa et al. [30] highlighted seasonal and anthropogenic influences on groundwater contamination in Nigeria.

Among chemical contaminants, PCBs remain particularly concerning due to their extreme toxicity and persistence. These compounds are widely distributed environmental pollutants capable of inducing mutagenic and carcinogenic effects in humans [8, 24, 25, 34, 39]. Exposure to dioxin-like PCBs has been associated with disruptions in liver vitamin balance, hepatotoxicity and impaired detoxification mechanisms, including oxidative stress, apoptotic activation and DNA damage [8, 25, 39]. Consequently, PCB contamination of drinking groundwater represents a significant human health concern.

Recent studies have increasingly documented PCB contamination in Nigerian groundwater. For instance, Ediagbonya et al. [10] assessed PCB risks in surface and groundwater across three Nigerian states, while Aralu et al. [5] conducted a systematic review of PCB levels and toxicity in Nigerian aquatic systems. Despite these existing studies, there remains a critical research gap regarding the integrated assessment of both microbial and chemical contaminants in the specific coastal aquifers of Ondo State. Therefore, this study aims to investigate the presence of fungi and polychlorinated biphenyls (PCBs) in groundwater from the coastal region of Ondo State, Nigeria, and to evaluate the associated human health risk from continued consumption of these groundwaters.

2 Methods

2.1 Study area and hydrogeological context

The study was conducted in coastal communities of Ondo State, Nigeria, specifically Ese-Odo, Irele and Ilaje Local Government Areas. The study area encompasses Igbobini and Irele and Omin in Ese-Odo and Irele LGAs respectively, and Igbokoda in Ilaje LGA. According to the 2006 population census, these areas have populations of 158,256 and 289,838 for Ese-Odo, Irele and Ilaje LGAs respectively [29]. The region is characterized by significant anthropogenic pressure, including the presence of small-scale mechanic workshops, e-waste recycling sites, and proximity to industrial zones involved in petroleum handling and electrical equipment maintenance, which serve as potential point sources for chemical contaminants such as PCBs.

The hydrogeological setting is characterized by shallow unconfined aquifers within the coastal plain sands of the Niger Delta Basin. Groundwater occurs at depths ranging from 5 to 30 m below ground level with seasonal fluctuations influenced by monsoonal rainfall patterns [33]. The aquifer materials consist primarily of fine to medium-grained sands with intermittent clay lenses that may influence contaminant transport and attenuation. Similar coastal aquifer systems in West Africa show vulnerability to anthropogenic contamination due to high permeability and limited natural attenuation capacity [6, 17]. The dry season sampling period, December 2023 to February 2024, was selected to minimize dilution effects from rainfall infiltration and to assess worst-case contaminant

concentrations in domestic water supplies [20]. Details regarding well characteristics for each sampled location are provided in Table S3 (Supplementary Material).

Sampling locations and coordinates were: Ese-Odo LGA, Market Street Igbobini (Lat $6^{\circ} 30' 4.59''$ N, Long $4^{\circ} 49' 20.976''$ E), Health Centre Igbobini ($6^{\circ} 30' 5.01''$ N, Long $4^{\circ} 49' 17.802''$ E) and Bestman Avenue ($6^{\circ} 30' 27.12''$ N, Long $4^{\circ} 49' 16.98''$ E). From Irele LGA, samples were collected from Seja Omin ($6^{\circ} 28' 45.156''$ N, Long $4^{\circ} 53' 17.7042''$ E), Ebute Road, Ode-Irele ($6^{\circ} 29' 30.492''$ N, Long $4^{\circ} 51' 50.743''$ E) and General Hospital, Ode-Irele ($6^{\circ} 29' 36.984''$ N, Long $4^{\circ} 51' 43.638''$ E). Ilaje LGA samples were drawn from Sunomo Igbokoda ($6^{\circ} 23' 11.136''$ N, Long $4^{\circ} 46' 12.966''$ E), Akins Motel, Igbokoda ($6^{\circ} 21' 21.546''$ N, Long $4^{\circ} 48' 3.942''$ E) and Olugbokoda Palace, Igbokoda ($6^{\circ} 21' 15.384''$ N, Long $4^{\circ} 47' 47.154''$ E). Figure 1 shows the map of the study area and sampling locations.

Table 1 summarizes the sampling locations and the number of samples collected from each site to clarify the distribution of the 50 samples across the nine locations. The unequal number of samples per location reflects the availability of functional domestic wells; this unbalanced design was accounted for in statistical analyses, and conclusions regarding groundwater quality are limited to dry-season conditions.

2.2 Sample collection and distribution

Fifty groundwater samples were collected from nine distinct locations during the dry season, December 2023 to February 2024. At each of the nine locations, replicate samples (5–6 per site) were collected from distinct hand-dug wells or boreholes used for domestic supply. These replicates represent biological replicates capturing spatial variability within each community, rather than technical replicates from a single source. Samples were collected using sterile 500 mL polypropylene bottles following standard

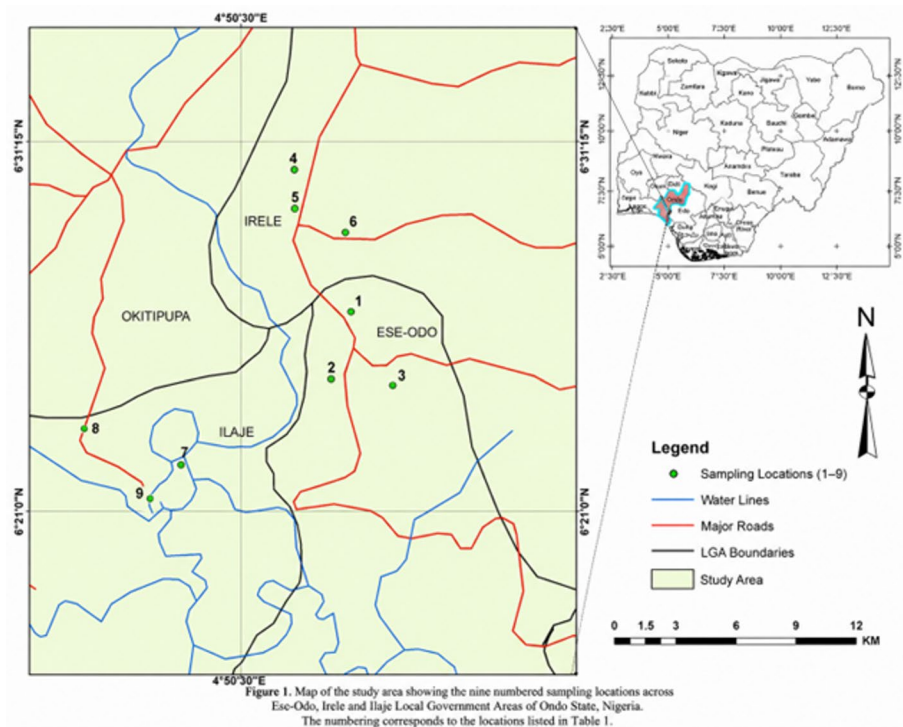


Fig. 1 Map of the study area showing the nine numbered sampling locations across Ese-Odo, Irele and Ilaje Local Government Areas of Ondo State, Nigeria. The numbering corresponds to the locations listed in Table 1

Table 1 Summary of sampling locations and sample distribution

No.	Location	LGA	Number of samples
1	Market Street, Igbobini	Ese-Odo	6
2	Health Centre, Igbobini	Ese-Odo	6
3	Bestman Avenue	Ese-Odo	5
4	Seja Omin	Irele	6
5	Ebute Road, Ode-Irele	Irele	6
6	General Hospital, Ode-Irele	Irele	5
7	Sunomo, Igbokoda	Ilaje	6
8	Akins Motel, Igbokoda	Ilaje	5
9	Olugbokoda Palace, Igbokoda	Ilaje	5
	Total		50

protocols for microbiological and chemical analysis [36]. For PCB analysis, two 500 mL bottles were combined to obtain a 1 L composite sample per replicate, ensuring sufficient volume for liquid-liquid extraction. All samples were stored at 4 °C and analyzed within 24 h of collection. Sampling campaigns employed aseptic procedures including flame-sterilized taps where possible and immediate capping to prevent external contamination [20].

2.3 Chemicals and instruments

All chemicals used were of analytical grade. Methylene chloride (HPLC grade, Batch No. MC2023-09), hexane (HPLC grade, Batch No. HX2023-11), sodium hydroxide (AR grade, Batch No. SH2023-04), and sulfuric acid (AR grade, Batch No. SA2023-02) were purchased from Sigma-Aldrich. PCB standard reference materials (Batch No. PCB-STD-2023) were obtained from AccuStandard. Potato Dextrose Agar (Hi Media, Batch No. PDA-HM-2023) was used for microbiological analysis.

Instrumentation included an Agilent 7820 A gas chromatograph coupled to a 5975 C inert mass spectrometer (Agilent Technologies, USA). An autoclave (Model: Prestige Deluxe, Make: Hawkins) was used for sterilization. Microbial incubation was performed in a membrane incubator (Model: MI-400, Make: Labtech). pH adjustments were monitored using a digital pH meter (Model: PH-100, Make: Hanna Instruments).

2.4 Microbiological analysis

Potato Dextrose Agar (39 g, Hi Media) was dissolved in 1000 mL distilled water, sterilized by autoclaving at 121 °C for 15 min at 1.1 kg/cm² pressure [19]. Each water sample was serially diluted to the 6th dilution and 1 mL aliquots were inoculated into sterile Petri dishes, overlaid with pre-sterilized PDA and incubated at 28 ± 2 °C for 5 days for fungal isolation. Pure cultures were obtained by repeated streaking on PDA. Fungal identification relied solely on morphological and cultural characteristics following standard mycological keys [14]. An experienced mycologist independently verified the morphological identifications. Consequently, all ecological and health interpretations are restricted to the genus level. Species-level identifications presented in the data are tentative and require molecular confirmation for definitive resolution. Future studies should incorporate molecular methods, for example ITS sequencing, to improve taxonomic accuracy [37].

2.5 PCB extraction and quantification

2.5.1 Extraction protocol

PCB extraction followed liquid-liquid extraction procedures adapted from US EPA Method 3510 C and validated for Nigerian environmental matrices [10, 22]. One liter of water sample at neutral pH was serially extracted with methylene chloride. pH was adjusted using 10 N sodium hydroxide or 1:1 sulfuric acid solution as required. Laboratory control samples and blanks were processed alongside field samples. One milliliter of 200 ng/mL surrogate working solution was added to procedure blanks, LCS and all samples. Extracts were concentrated using Kuderna-Danish apparatus and exchanged into hexane (60 mL) for cleanup. The combined extract was concentrated to 1 mL final volume prior to GC-MS analysis [11].

2.5.2 GC-MS analysis

Analysis employed an Agilent 7820 A gas chromatograph coupled to a 5975 C inert mass spectrometer with an electron-impact source and triple-axis detector (Agilent Technologies). Separation used an HP-5 capillary column (30 m x 0.32 mm x 0.25 μ m film thickness) coated with 5% phenyl methyl siloxane. Helium carrier gas flowed at 1.2 mL/min constant flow with an initial pressure of 0.26 psi and an average velocity of 40.00 cm/sec. Injection volume was 1 μ L in splitless mode at 250 °C. Oven program: 110 °C hold 1 min then ramp to 310 °C at 15 °C/min hold 3 min. The total run time for the analysis was 17.33 min. Solvent delay: 3 min. Mass spectrometer operated at 70 eV in EI mode with ion source 230 °C, quadrupole 150 °C, transfer line 300 °C. Data acquisition used SIM and scan modes (m/z 35–550 at 2.0 scans/sec). Calibration was performed using a seven-point calibration curve ranging from 10 ng/L to 500 ng/L, with a coefficient of determination (R^2) > 0.99 for all congeners.

Although MRM parameters were requested, the analytical method employed standard GC-MS in electron impact (EI) mode with selected ion monitoring (SIM). Therefore, MRM transitions and collision energies are not applicable. Instead, Table 2 provides the precursor ions and target SIM ions used for quantification.

2.5.3 Quality control

Method detection limits (LOD) ranged from 0.03 to 1.2 ng/L, and limits of quantification (LOQ) ranged from 0.10 to 4.0 ng/L, determined at signal-to-noise ratios of 3 and 10, respectively. Field blanks and spiked blanks were analyzed to assess background contamination. Procedure and solvent blanks contained no target compounds. Surrogate recoveries ranged from 83 to 112%, meeting acceptance criteria established for PCB analysis in West African environmental samples [10, 22]. Congener-specific LOD

Table 2 GC-MS selected ion monitoring (SIM) parameters for analyzed PCB congeners

Congener	Precursor ion (m/z)	Target SIM ions (m/z)	Dwell time (ms)	Ionization energy (eV)
PCB1	152	152, 154	20	70
PCB6	186	186, 188	20	70
PCB18	220	220, 222	20	70
PCB31	254	254, 256	20	70
PCB44	288	288, 290	20	70
PCB153	360	360, 362	20	70
PCB180	394	394, 396	20	70

Full SIM parameters for all 18 congeners followed the same EI fragmentation patterns at 70 eV

(0.03–0.09 ng/L) and LOQ (0.10–0.28 ng/L) were determined independently from the seven-point calibration curve using signal-to-noise ratios of 3 and 10 applied to chromatographic peak responses in the original water matrix (no extraction/concentration factor was applied; Supplementary Table S1). Because instrumental response is linear below the lowest calibration standard, congener concentrations between LOD and LOQ are reported as semi-quantitative trace-level detections rather than fully quantified values; this applies to most individual congeners other than PCB18, PCB31 and PCB44, which exceeded their respective LOQs at several locations (Table 5). Calibration statistics and congener-specific recovery rates are reported in Supplementary Table S1. This analytical approach follows protocols employed by Iwegbue et al. [22] and Ediabonyia et al. [11] for PCB analysis in aqueous matrices from the Niger Delta region. Non-detects were substituted with half the limit of detection (LOD/2) for statistical analysis. This approach is a widely accepted standard practice in environmental risk assessment to provide a conservative estimate of exposure when non-detects constitute less than 50% of the dataset, although more robust censored-data statistical approaches exist.

2.6 Data analysis

Data were organized, cleaned and checked for accuracy using IBM SPSS Statistics v28.0. Prior to parametric testing, normality was assessed using the Shapiro-Wilk test, and homogeneity of variance was evaluated using Levene's test. Shapiro-Wilk ($p > 0.05$) and Levene's tests ($p > 0.05$) confirmed that the data met the assumptions of normality and homogeneity of variance, respectively. ANOVA was employed to assess spatial variation, followed by Tukey HSD post hoc tests for pairwise comparisons, as data met assumptions of normality and homogeneity of variance. Significance level was set at $p < 0.05$. Descriptive statistics (mean, interquartile range) summarized PCB concentrations across sites. Similar statistical approaches have been applied in groundwater quality assessments in coastal Nigeria [9, 20]. At the location level (9 locations), non-detects (BDL) were most frequent for PCB52 and the Tri-PCB homologue group (5 of 9 locations, 55.6%), and occurred at 1–2 of 9 locations (11.1–22.2%) for PCB187, PCB183, PCB180, PCB170, PCB206, Hepta-PCB and Octa-PCB; all other congeners had no location-level non-detects. Location-level, rather than full replicate-level, non-detect frequencies are reported here because raw replicate-by-replicate concentration data were not retained in a form permitting a formal censored-data sensitivity analysis; LOD/2 substitution followed by ANOVA/Tukey HSD was applied uniformly, and a non-parametric Kruskal-Wallis comparison (reported for the fungal colony-count data) was not additionally performed for every PCB congener. No correction for multiple comparisons (e.g., Bonferroni or false-discovery-rate adjustment) was applied across the congener- and homologue-level ANOVA tests; given the number of comparisons, the reported p-values should be interpreted with this in mind. Statistical comparisons of fungal occurrence and colony counts among locations were performed using the Kruskal-Wallis test due to non-normal distribution of the microbiological data.

2.7 Human health risk assessment

Risk assessment evaluated human exposure to measured PCB congeners in groundwater through ingestion and dermal contact pathways. An inhalation pathway does not appear in the results because a groundwater-to-air volatilization model was not developed for

this dataset (see Sect. 2.7.1). Exposure-point concentrations used for risk calculation were arithmetic mean congener concentrations at each location as reported in Table 5. The 95% upper confidence limit approach originally specified in the formula below was not applied since it requires replicate-level concentration data that were not retained in a form permitting recalculation. The CUCL equation is retained below for methodological completeness and as the recommended approach for future studies with accessible replicate-level data:

$$C_{UCL} = \bar{X} + Z \times \frac{S}{n^{0.5}} \times \left(1 + \frac{Z^2}{2n} + \frac{\beta}{6n} \right)^{0.5} \quad (1)$$

Where:

C_{UCL} = Upper confidence limit concentration.

$\bar{X}$ = Arithmetic mean concentration.

S = Standard deviation.

Z = 95th percentile of the standard normal distribution (1.645).

n = Sample number.

β = Skewness.

Ediabonya et al. [10], Iwegbue [22].

Carcinogenic risk assessment was not conducted because the PCB congeners required for cancer risk estimation were not included in the analytical determination. Only non-carcinogenic risk, specifically Hazard Quotient and Hazard Index, was evaluated consistent with the available data.

2.7.1 Exposure parameters

Chronic daily intake for non-carcinogenic effects was calculated for two exposure pathways, namely ingestion and dermal contact. The inhalation-pathway equation is retained below for completeness but was not used to generate the results in Tables 7 and 8 since no groundwater-to-air volatilization model was applied. In the equations below, C_i represents the arithmetic mean concentration of the specific congener at the sampling location.

Ingestion Pathway (Eq. 2)

$$CDI_{ing-nc} = \frac{C_i \times IngR \times EF \times ED}{BW \times AT} \quad (2)$$

Inhalation Pathway (Eq. 3) [Not used in calculation]

$$CDI_{inh-nc} = \frac{C_i \times InhR \times EF \times ET \times ED}{PEF \times 24 \times BW \times AT} \quad (3)$$

Dermal Contact Pathway (Eq. 4)

$$CDI_{derm-nc} = \frac{C_i \times SA \times AF \times ABS_d \times EF \times ED}{BW \times AT} \quad (4)$$

Hazard Quotient (HQ) and Hazard Index (HI).

The hazard quotient for each exposure route was calculated using:

$$HQ = \frac{CDI}{RfD} \quad (5)$$

The total non-carcinogenic hazard index was estimated using the sum of the ingestion and dermal hazard quotients:

$$HI = HQ = HQ_{ing} + HQ_{derm} \quad (6)$$

Where:

RfD = Reference dose obtained from ATSDR guidance on PCB health effects [3].

CDI = Chronic daily intake for each exposure pathway.

All exposure parameters including toxicity reference values are defined in Table 3. Adult water ingestion rate was set at 2.0 L/day consistent with exposure factors applied in PCB risk assessments for Nigerian populations [10, 22]. Child ingestion rate was 1.0 L/day. Body weights were 70 kg for adults and 15 kg for children. PCB toxicity reference values were derived from ATSDR guidance on PCB health effects [3].

2.7.2 Unit consistency and population specification

PCB concentrations measured in ng/L were explicitly converted to mg/kg by multiplying by 1×10^{-6} , assuming a water density of 1 kg/L. This corrects previous unit conversion errors where ng/L was erroneously treated as mg/L. Risk calculations were performed separately for adult (70 kg) and child (15 kg) exposure scenarios. Two complementary risk metrics are presented: (i) Tables 7 and 8 report the maximum single-congener hazard quotient (HQ), which is higher for adults (0.187, PCB31) than children (0.005, PCB31) under the exposure parameters applied to that specific congener; and (ii) Supplementary Table S4 reports the cumulative, all-congener hazard index (HI) per location, which is consistently higher for children than adults (approximately 2.1–2.4-fold) at every sampling location, consistent with children's greater ingestion rate per unit body weight. The cumulative HI in Supplementary Table S4 is the more appropriate metric for characterizing overall non-carcinogenic risk from combined PCB exposure across all congeners, whereas the single-congener maxima in Tables 7 and 8 illustrate which

Table 3 Risk assessment exposure parameters, toxicity reference values, and constants

Parameter	Definition	Value used	Source
BW	Body weight	70 kg (adult), 15 kg (child)	[3, 22]
IngR	Ingestion rate	2.0 L/day (adult), 1.0 L/day (child)	[3, 10]
EF	Exposure frequency	350 days/year	Assumed
ED	Exposure duration	30 years (adult), 6 years (child)	[3, 22]
AT_nc	Averaging time (non-carcinogen)	ED $\times$ 365 days	Calculation
AT_c	Averaging time (carcinogen)	70 years $\times$ 365 days	ATSDR [3]
SA	Skin surface area	18,000 cm ² (adult), 6,600 cm ² (child)	Iwegbue et al. [22]; Ediagbonya et al. [10]
AF	Skin adherence factor	0.07 mg/cm ² -event	ATSDR [3]
ABS_d	Dermal absorption factor	0.14 for PCBs	[3, 22]
PEF	Particle emission factor	Not applicable (inhalation pathway excluded)	N/A
ET	Exposure time	0.58 h/day (showering)	Assumed
RfD	Oral reference dose (non-carcinogenic)	2.0×10^{-5} mg/kg/day	ATSDR [3]
CSF	Oral cancer slope factor (carcinogenic)	Not applicable (carcinogenic risk not evaluated)	N/A
CF	Conversion factor	1×10^{-6} (ng/L to mg/kg)	Calculation

individual congener contributes most to that risk. Background PCB exposure levels in Nigerian populations suggest cumulative risk considerations are warranted [16]. It is important to note that while the PCB health risk assessment was quantitative, the fungal health risk assessment was qualitative and inferred from organism occurrence rather than calculated exposure assessment.

3 Results and discussion

3.1 Fungal contaminants in groundwater

The study revealed the presence of twelve fungal isolates belonging to eight genera in the assayed groundwater samples from the coastal region of Ondo State, Nigeria (Table 4). *Rhodotorula* spp. was the most frequently detected genus (77.8% of samples), followed by *Rhizopus* spp. (66.7%) and *Fusarium* spp. (55.6%). *Aspergillus* spp., a genus containing known opportunistic pathogens, was detected in 55.6% of samples. Other genera included *Mucor* spp. (44.4%), *Trichophyton* spp. (33.3%), *Penicillium* spp. (22.2%), and *Torulopsis* spp. (11.1%).

Among these isolates, *Torulopsis candida* and *Rhodotorula rubra* are yeasts while the six other genera are molds. This result corroborates the report of Hageskal et al. [14] that fungi obtained from drinking water include yeast but mostly a variety of filamentous molds including those with potential for disease causation, allergy induction and toxigenic properties. Among the molds recovered were *Penicillium* species, *Aspergillus* species and *Fusarium* species which are known to produce mycotoxins such as patulin, aflatoxins and zearalenone even if in diluted or less concentrated amounts. Although these genera are known to produce mycotoxins, actual mycotoxin production was not measured in this study and is highly dependent on specific environmental conditions. Therefore, the potential health risk is inferred from the presence of these genera rather than confirmed mycotoxin levels. These mycotoxins and metabolites may bioaccumulate with continued consumption over a long period and become harmful to human health.

Quantitative analysis revealed that median fungal colony counts ranged from 15 to 45 CFU/100 mL across the sampling locations. These counts are reported here descriptively; no universally applicable WHO or Nigerian regulatory numerical limit for fungal counts in drinking water was identified, and no single benchmark value is invoked to characterize the water as contaminated. Values represent medians with interquartile ranges provided in Supplementary Table S2; $n = 5$ –6 replicates per location. The Kruskal-Wallis test indicated a statistically significant difference in fungal colony counts among the nine locations ($H(8) = 45.54$, $p < 0.001$), confirming spatial heterogeneity in fungal contamination. The prevalence of *Aspergillus* and *Fusarium* species aligns with reports from groundwater systems in Ghana [4] but contrasts with lower detection rates in treated urban supplies in South Africa [26]. The presence of genera capable of mycotoxin production in some environmental contexts underscores the need for mycotoxin screening and molecular confirmation in future assessments; pathogenicity, viability and toxin production were not directly tested in this study. The detection of these fungal genera in untreated groundwater warrants microbiological monitoring and further investigation of their potential public-health significance, consistent with the occurrence-level findings of Ikuesan and Ediagbonya [20]. The percentage occurrence of fungal genera in the assayed groundwater samples is presented in Fig. 2, which illustrates the dominance of *Rhodotorula*, *Rhizopus* and *Aspergillus* species across the sampled locations

Table 4 Identity and distribution of fungal isolates in water samples from the coastal environment of Ondo State, Nigeria

Genus	Species	IR-E	IR-GH	IR-O	GB-S	GB-M	GB-P	IG-BM	IG-HC	IG-MR	Total (%)
<i>Fusarium</i>	<i>F. verticillioides</i>	+	+	-	+	+	-	+	-	-	5 (55.6)
<i>Rhizopus</i>	<i>R. stolonifer</i>	-	+	-	+	+	+	+	+	-	6 (66.7)
<i>Mucor</i>	<i>M. hiemalis</i>	-	-	+	+	-	+	-	+	-	4 (44.4)
<i>Rhodotorula</i>	<i>R. rubra</i>	+	+	+	+	+	+	+	-	-	7 (77.8)
<i>Trichophyton</i>	<i>T. quinckeanum</i>	-	+	-	-	-	-	-	-	-	1 (11.1)
<i>Aspergillus</i>	<i>A. fumigatus</i>	-	-	+	-	+	-	+	-	+	5 (55.6)
<i>Penicillium</i>	<i>Penicillium</i> spp.	-	-	-	-	-	-	-	+	+	2 (22.2)
<i>Fusarium</i>	<i>F. solani</i>	-	-	-	-	-	-	+	-	-	1 (11.1)
<i>Trichophyton</i>	<i>T. rubrum</i>	+	-	-	+	-	+	-	-	-	3 (33.3)
<i>Aspergillus</i>	<i>A. niger</i>	-	+	+	-	-	-	-	-	+	3 (33.3)
<i>Penicillium</i>	<i>P. chrysogenum</i>	-	-	+	-	-	-	-	-	-	1 (11.1)
<i>Torulopsis</i>	<i>T. candida</i>	-	-	-	-	-	+	-	-	-	1 (11.1)
Total		3	5	5	5	4	5	5	3	3	

Location codes: IR-E, IR-GH, IR-O = Irele LGA; GB-S, GB-M, GB-P = Ilaje LGA; IG-HC, IG-BM, IG-MR = Ese-Odo LGA. + = present, - = absent Species-level entries in this column are tentative morphological identifications only and are not molecularly confirmed; they are provided for reference and are not treated as confirmed taxonomic identities anywhere in the Abstract, Results, Discussion or Conclusion, all of which restrict interpretation to the genus level. Molecular confirmation (e.g., ITS sequencing) is recommended for future studies

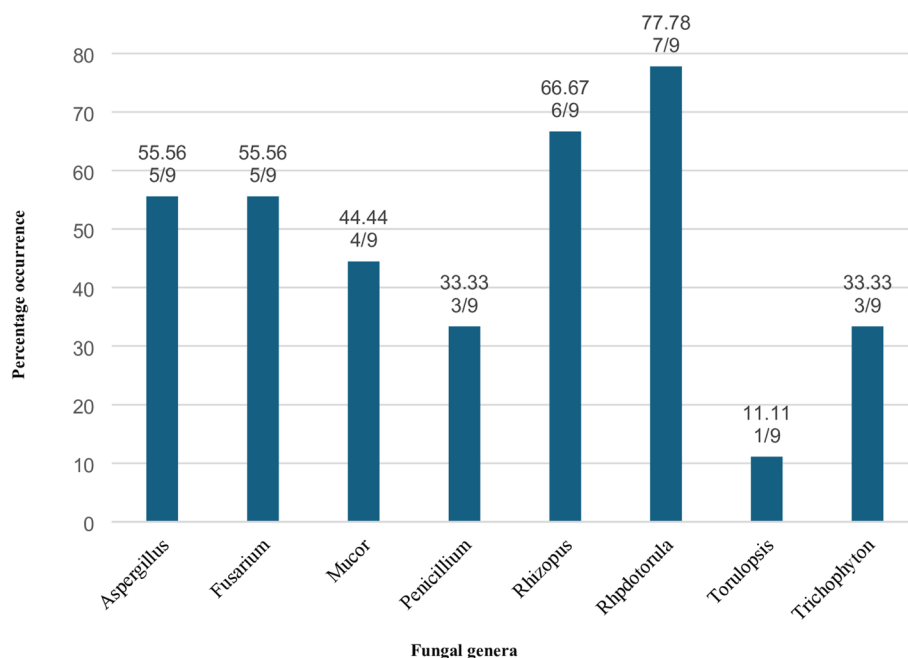


Fig. 2 Percentage occurrence of fungal genera in assayed groundwater samples from coastal Ondo State, Nigeria in the coastal region of Ondo State. Figure 2 reports the number of wells/samples (n) underlying each percentage; see the Fig. 2 caption for the exact denominator and interval method used.

Values represent the proportion of sampling locations ($n = 9$ distinct wells) where each genus was detected. Raw detection counts (e.g., 7/9) are displayed above each bar to indicate sample size. No confidence intervals are presented due to the small number of locations. Species-level identifications are tentative and based on morphology only, all interpretations are restricted to genus level.

3.2 PCB congener distribution and concentrations

Table 5 shows the concentrations of polychlorinated biphenyl congeners across the nine sampling locations. PCB concentrations ranged from below the detection limit (BDL) to 3.90 ± 0.03 ng/L for PCB18 in IGHC samples, the highest single-congener value recorded; estimated Σ PCB (sum of congener means) ranged from approximately 0.25 to 10.0 ng/L across locations (see Sect. 3.3). The degree of chlorination significantly influences PCB persistence and mobility in aquatic environments. According to Grimm et al. [13], PCBs with low chlorination frequently metabolize faster than those with high chlorination, though they may exhibit greater mobility in groundwater systems.

The investigation revealed a variety of PCB congeners, from mono- to octa-chlorinated biphenyls, with significant variations between locations ($p < 0.05$). One of the most remarkable findings was the elevated concentration of various PCBs, especially in the IGHC samples. PCB6, a dichlorinated biphenyl, varied significantly between locations ($F = 20.329$, $p < 0.001$), with IGHC having the highest concentration (0.07 ± 0.01 ng/L), approximately 3.5 times greater than other locations. PCB31, a trichlorinated biphenyl, was considerably higher in both IGHC (3.00 ± 0.10 ng/L) and IRGH (3.39 ± 0.34 ng/L) samples, indicating substantial elevation compared to other locations ($F = 353.824$, $p < 0.001$).

Table 5 Concentration of polychlorinated biphenyl (PCB) congeners in various locations (ng/L)

Congener	IGGH	IGHC	IRE	IRGH	IRO	GBM	GBP	GBS	IGBM	F	Sig
PCB1	0.02 ± 0.01 ^{ab}	0.02 ± 0.01 ^{ab}	0.02 ± 0.01 ^{ab}	0.02 ± 0.01 ^{ab}	0.02 ± 0.01 ^{ab}	0.00 ± 0.00 ^b	0.01 ± 0.00 ^{ab}	0.00 ± 0.00 ^b	0.00 ± 0.00 ^b	4.031	0.007
PCB6	0.02 ± 0.01 ^a	0.07 ± 0.01 ^b	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.01 ± 0.00 ^a	0.01 ± 0.00 ^a	0.01 ± 0.00 ^a	0.01 ± 0.00 ^a	20.329	0.000
PCB18	0.01 ± 0.01	3.90 ± 0.03	0.01 ± 0.01	0.50 ± 0.03	0.51 ± 0.01	1.37 ± 2.37	1.38 ± 2.36	0.01 ± 0.00	1.37 ± 2.37	2.467	0.053
PCB31	1.25 ± 0.02 ^a	3.00 ± 0.10 ^b	0.01 ± 0.01 ^c	3.39 ± 0.34 ^b	0.02 ± 0.01 ^c	0.16 ± 0.00 ^c	0.08 ± 0.08 ^c	0.06 ± 0.09 ^c	0.03 ± 0.02 ^c	353.82	0.000
PCB52	BDL	BDL	BDL	BDL	BDL	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.397	0.759
PCB44	0.01 ± 0.01 ^a	2.50 ± 0.04 ^b	0.01 ± 0.01 ^a	0.10 ± 0.01 ^c	0.03 ± 0.01 ^a	0.01 ± 0.01 ^a	0.64 ± 1.09 ^d	0.01 ± 0.01 ^a	0.93 ± 0.94 ^d	9.000	0.000
PCB66	0.00 ± 0.01	0.01 ± 0.01	0.00 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.980	0.482
PCB99	0.01 ± 0.01 ^a	0.05 ± 0.01 ^b	0.02 ± 0.01 ^a	0.03 ± 0.01 ^{ab}	0.02 ± 0.01 ^a	0.01 ± 0.01 ^a	0.01 ± 0.00 ^a	0.01 ± 0.01 ^a	0.01 ± 0.00 ^a	8.126	0.000
PCB87	0.02 ± 0.01 ^a	0.12 ± 0.01 ^b	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.03 ± 0.01 ^a	0.05 ± 0.05 ^a	0.08 ± 0.05 ^a	0.05 ± 0.05 ^a	0.01 ± 0.00 ^a	3.777	0.009
PCB109	0.00 ± 0.01 ^a	0.11 ± 0.01 ^b	0.00 ± 0.01 ^a	0.03 ± 0.01 ^b	0.02 ± 0.01 ^a	0.03 ± 0.04 ^b	0.01 ± 0.00 ^a	0.01 ± 0.00 ^a	0.03 ± 0.04 ^b	9.613	0.000
PCB151	0.00 ± 0.01 ^a	0.03 ± 0.01 ^b	0.00 ± 0.01 ^a	0.01 ± 0.01 ^{ab}	0.01 ± 0.01 ^{ab}	0.01 ± 0.00 ^{ab}	0.00 ± 0.00 ^{ab}	0.00 ± 0.00 ^{ab}	0.01 ± 0.00 ^{ab}	5.214	0.002
PCB153	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.03 ± 0.01 ^a	0.02 ± 0.01 ^a	0.01 ± 0.00 ^a	0.01 ± 0.00 ^a	0.01 ± 0.00 ^a	0.01 ± 0.00 ^a	3.538	0.012
PCB146	0.02 ± 0.02 ^a	0.02 ± 0.01 ^a	0.03 ± 0.01 ^a	0.02 ± 0.01 ^a	0.03 ± 0.01 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	6.837	0.000
PCB137	0.02 ± 0.02 ^a	0.03 ± 0.01 ^b	0.03 ± 0.01 ^b	0.01 ± 0.01 ^a	0.01 ± 0.01 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.01 ± 0.00 ^a	5.256	0.002
PCB187	BDL	0.01 ± 0.01 ^a	0.09 ± 0.01 ^b	0.01 ± 0.01 ^a	0.02 ± 0.01 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	54.323	0.000
PCB183	BDL	0.02 ± 0.01 ^a	BDL	0.02 ± 0.01 ^a	0.03 ± 0.01 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.01 ± 0.02 ^a	0.00 ± 0.00 ^a	4.447	0.010
PCB180	BDL	0.02 ± 0.01 ^a	BDL	0.03 ± 0.01 ^a	0.04 ± 0.01 ^a	0.03 ± 0.01 ^a	0.01 ± 0.00 ^a	0.03 ± 0.00 ^a	0.03 ± 0.01 ^a	3.088	0.038
PCB170	BDL	0.02 ± 0.01 ^a	BDL	0.04 ± 0.01 ^b	0.05 ± 0.01 ^b	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	27.377	0.000
PCB206	BDL	0.09 ± 0.01 ^b	BDL	0.04 ± 0.01 ^b	0.04 ± 0.01 ^b	0.04 ± 0.02 ^b	0.04 ± 0.01 ^b	0.04 ± 0.01 ^b	0.03 ± 0.01 ^b	7.159	0.001
Mono-PCB	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.02 ± 0.01 ^a	0.00 ± 0.00 ^a	0.01 ± 0.00 ^a	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a	4.031	0.007
Di-PCB	0.03 ± 0.02 ^a	3.97 ± 0.04 ^b	0.03 ± 0.02 ^a	0.52 ± 0.04 ^a	0.53 ± 0.02 ^a	1.38 ± 2.37 ^a	1.38 ± 2.36 ^a	0.02 ± 0.00 ^a	1.38 ± 2.37 ^a	2.525	0.049
Tri-PCB	BDL	BDL	BDL	BDL	BDL	0.18 ± 0.01	0.72 ± 1.06	0.08 ± 0.08	0.96 ± 0.94	1.072	0.414
Tetra-PCB	0.00 ± 0.01	0.01 ± 0.01	0.00 ± 0.01	0.01 ± 0.01	0.01 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.980	0.482
Penta-PCB	0.03 ± 0.02 ^a	0.17 ± 0.02 ^a	0.04 ± 0.02 ^a	0.05 ± 0.02 ^a	0.05 ± 0.02 ^a	0.06 ± 0.05 ^a	0.09 ± 0.05 ^a	0.06 ± 0.06 ^a	0.02 ± 0.00 ^a	4.964	0.002
Hexa-PCB	0.05 ± 0.03 ^a	0.07 ± 0.03 ^a	0.05 ± 0.03 ^a	0.06 ± 0.03 ^a	0.06 ± 0.03 ^a	0.02 ± 0.00 ^a	0.02 ± 0.00 ^a	0.01 ± 0.00 ^a	0.01 ± 0.00 ^a	3.585	0.012
Hepta-PCB	BDL	0.07 ± 0.04 ^a	BDL	0.10 ± 0.04 ^a	0.14 ± 0.04 ^a	0.03 ± 0.01 ^a	0.02 ± 0.01 ^a	0.05 ± 0.02 ^a	0.03 ± 0.01 ^a	7.092	0.001
Octa-PCB	BDL	0.09 ± 0.01 ^b	BDL	0.04 ± 0.01 ^a	0.04 ± 0.01 ^a	0.04 ± 0.02 ^a	0.04 ± 0.01 ^a	0.04 ± 0.01 ^a	0.03 ± 0.01 ^a	7.159	0.001

BDL = below detection limit (LOD); values reported as BDL were below LOD and were not included as quantified concentrations. This is distinct from values that were quantified but fell below the limit of quantification (LOQ) — see Sect. 3.4 note on LOQ reconciliation. Values are mean ± SD (n = 5–6 replicates per location). Values with different superscripts within a row differ significantly ($p < 0.05$, Tukey HSD). Location codes: IGGH = Igboini General Hospital, IGHC = Igboini Health Centre, IRE = Irele, IRGH = Irele General Hospital, IRO = Ode-Irele, GBM = Igboini Bestman, GBP = Igboini Palace, GBS = Igboini Sunomo, IGBM = Igboini Market

High standard deviations were observed for certain values, such as 1.37 ± 2.37 ng/L for PCB18 in GBM, reflect spatial heterogeneity in PCB distribution within sampling locations. Non-detect values were substituted with LOD/2 for statistical analysis. This variability suggests point sources or differential transport mechanisms affecting congener distribution.

This investigation also revealed interesting patterns among PCB groupings. Dichlorinated biphenyls differed marginally ($F = 2.525$, $p = 0.049$), whereas penta- ($F = 4.964$, $p = 0.002$), hepta- ($F = 7.092$, $p = 0.001$), and octa-chlorinated ($F = 7.159$, $p = 0.001$) biphenyls varied significantly between locations. This indicates that the degree of chlorination may influence how these chemicals accumulate or persist in different environments. However, not all PCBs followed this pattern. Some, such as PCB52 and PCB66, revealed no significant variations between locations ($p > 0.05$), implying more uniform distribution or possibly reflecting background contamination levels. It is worth noting that numerous PCBs were below the detection limit in several locations, particularly IGGH samples. For example, PCB187, PCB183, PCB180, PCB170, and PCB206 were all below the detection limits in IGGH samples.

Statistical reporting was verified, and superscript notation for post-hoc comparisons was corrected. Superscripts were removed for congeners where the ANOVA F-statistic was non-significant ($p > 0.05$), specifically PCB18 ($p = 0.053$), PCB52, PCB66, Tri-PCBs, and Tetra-PCBs, to ensure internal consistency.

Figure 3 illustrates the spatial variation in the relative contribution of PCB homologue groups across the nine sampling locations. Overall, lower-chlorinated homologues, particularly Di-PCBs, constituted a substantial proportion of the total PCB burden at most locations, contributing on average approximately 52% (range ~ 8–90% across the nine locations, calculated from the homologue-group means in Table 5) of the summed homologue concentration; this proportion varied considerably by location and was not uniformly dominant. This pattern is consistent with the greater environmental mobility and aqueous transport of lower-chlorinated congeners, which are more likely to migrate through permeable sandy aquifer materials than highly chlorinated congeners [13].

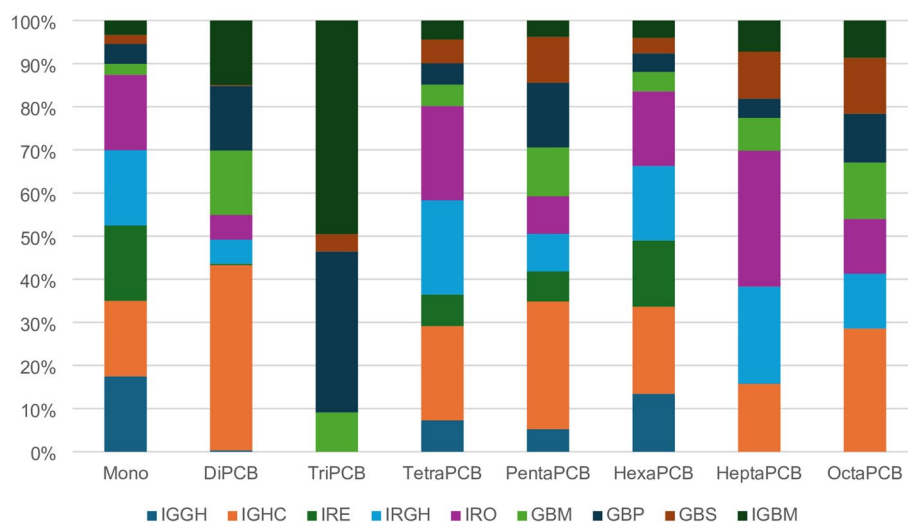


Fig. 3 Relative contribution of PCB homologues (Mono- to Octa-chlorinated) across nine sampling locations in coastal Ondo State. Values represent the percentage contribution of each homologue group to the total PCB concentration at each sampling location ($n = 5-6$ per location)

In contrast, some locations exhibited relatively greater contributions from higher-chlorinated homologues, including Hexa-, Hepta-, and Octa-PCBs. The elevated presence of these congeners at the Igbokoda Health Centre (IGHC) suggests the possible influence of localized historical PCB sources or enhanced retention within fine-grained sediment fractions due to the stronger sorption characteristics of highly chlorinated compounds, though sorption coefficients and sediment characteristics were not directly measured. Possible sources could include legacy electrical equipment containing PCB-based dielectric fluids or improper disposal of transformer materials; these are plausible explanations rather than determined sources, as formal source apportionment was not performed.

The distribution of PCB homologues also varied considerably among sampling locations, indicating that PCB contamination is spatially heterogeneous. While some locations were characterized by a predominance of lower-chlorinated homologues, others showed a more balanced contribution from both lower- and higher-chlorinated groups. Such differences may reflect variations in contamination sources, environmental transport processes, weathering and sediment-water partitioning, though these processes were not directly measured. Overall, the observed homologue profiles are consistent with a combination of recent and legacy PCB inputs together with site-specific hydro-geochemical conditions, though this interpretation should be regarded as a plausible explanation rather than a demonstrated source determination, since formal source apportionment was not conducted.

3.3 Comparison with regional and global studies

In this investigation, lower quantities of PCBs were detected in water samples compared to larger concentrations recorded in other sites in Nigeria and Western countries, as shown in Table 6. It should be clarified that 3.90 ng/L (PCB18 at IGHG) was the highest individual-congener concentration, not the total PCB concentration. Summing the mean concentrations of all 18 measured congeners at each location (Table 5) gives an estimated total (Σ PCB) concentration ranging from approximately 0.25 ng/L (GBS) to

Table 6 Comparison of PCB concentrations in this study with other studies

Congener	Present Study (ng/L) Range	Eze et al. [12] River Otamiri (ng/L) Range	Oregel-Zamudio et al. [32] Tropical Lake (ng/L) Range
PCB44	0.01–2.50	24.47–40.39	2250–2270
PCB1	0.00–0.02	ND	2140–2500
PCB18	0.01–1.38	29.12–36.16	2490–2750
PCB52	BDL – 0.00	1.32–7.03	1940–2320
PCB66	0.00–0.01	0.87–4.08	2320–2360
PCB99	0.00–0.05	ND	1690–1890
PCB87	0.02–0.10	ND	1980–2530
PCB146	0.00–0.03	ND	1870–1890
PCB153	0.00–0.03	38.30–50.93	2040–2490
PCB187	0.00–0.09	3.90–7.13	1980–2080
PCB180	0.02–0.03	12.56–33.79	1750–1980
PCB170	0.00–0.04	28.59–47.38	ND
PCB206	0.03–0.09	17.98–39.42	1590–2270

Oregel-Zamudio et al. [32] values were originally reported in ng/mL. To ensure unit compatibility, these values have been converted to ng/L by multiplying by 1000

BDL, below detection limit; ND, not determined

approximately 10.0 ng/L (IGHC); this location-mean-based estimate is provided for transparency, and a fully rigorous Σ PCB range calculated from individual replicate sums, rather than location means, should be reported by the authors from the raw dataset. Even at its highest estimated value, Σ PCB remains well below the U.S. EPA maximum allowable limit of 500 ng/L for total PCBs in drinking water, indicating that chemical contamination levels do not currently exceed this regulatory threshold. This difference is attributed to fewer anthropogenic activities in the study region. The number of lower chlorinated PCBs (PCB1, PCB6, PCB18, PCB31, PCB52, PCB44, PCB66, PCB99, PCB87, PCB109) detected was comparable to the number of higher chlorinated PCBs (PCB151, PCB153, PCB146, PCB137, PCB187, PCB183, PCB180, PCB170, PCB206).

A previous assessment examining PCB levels across the nation in various environmental media indicated concentrations above the WHO maximum limit [2, 7, 18, 34]. PCBs can degrade or disintegrate in the environment, although this process is highly dependent on the chemical composition of the PCBs. The degradation process is also influenced by environmental conditions. PCBs are typically destroyed in the environment by either sunlight or microbial activity. Sunlight contributes to the breakdown of PCBs in the air, shallow water, and surface soils [15, 40]. PCBs are biodegraded by bacteria, algae, and fungi [41].

3.4 Human health risk assessment

Tables 7 and 8 present the calculated risk metrics for various PCB congeners, including Average Daily Dose (ADD), Chronic Daily Intake (CDI) values for ingestion (CDI_ing) and dermal contact (CDI_derm), and the associated Hazard Quotients (HQ) for both exposure routes. The CDI values indicate estimated daily intake levels, and the HQ values represent the potential risk of adverse non-cancer effects. The cumulative Hazard Index (HI) reflects the combined risks from all exposure routes.

Tables 7 and 8 present the corrected risk metrics for various PCB congeners. Previous calculations contained a unit conversion error where ng/L concentrations were erroneously treated as mg/L, leading to massive overestimations of risk. The corrected hazard quotient (HQ) values for both adults and children were found to be significantly below the threshold of 1.0, with the maximum adult HQ being 0.187 for PCB31 and the maximum child HQ being 0.005 for PCB31. These results indicate that the non-carcinogenic health risks associated with PCB exposure via groundwater ingestion and dermal contact are within acceptable limits. As clarified in Sect. 2.7, the C_i values in Table 7 (e.g., 0.9853 ng/L for PCB18, 1.534 ng/L for PCB31) are arithmetic mean concentrations at each location (Table 5), not 95% UCL values; the Methods have been revised accordingly to describe the mean-based exposure-point concentration approach actually used, since replicate-level data needed for a rigorous CUCL calculation were not available.

The values in the Tables 7 and 8 highlight the variability in risks among different PCB congeners, where some exhibit higher potential risks for non-cancer effects compared to others, though all remain safely below the threshold of 1.0. Overall, these results underscore the importance of considering both ingestion and dermal contact pathways when evaluating the potential health hazards linked to exposure to particular PCB contaminants.

Table 7 Evaluation of average daily exposure and hazard indices of PCBs for adult exposure scenario (70 kg body weight)

Congener	Concentration C _i (ng/L)	CDI _{ing} (mg/kg/day)	HQ _{ing}	CDI _{derm} (mg/kg/day)	HQ _{derm}	CDI _{total} (mg/kg/day)	HQ _{total}	HI
PCB1	0.02	5.48×10^{-10}	0.00027	4.83×10^{-8}	0.002416	4.89×10^{-8}	0.002444	0.002444
PCB6	0.03	8.22×10^{-10}	0.00041	7.25×10^{-8}	0.003625	7.33×10^{-8}	0.003666	0.003666
PCB18	0.9853	2.70×10^{-8}	0.001350	2.38×10^{-6}	0.119046	2.41×10^{-6}	0.120396	0.120396
PCB31	1.534	4.20×10^{-8}	0.002101	3.71×10^{-6}	0.185341	3.75×10^{-6}	0.187442	0.187442
PCB44	0.53	1.45×10^{-8}	0.000726	1.28×10^{-6}	0.064102	1.30×10^{-6}	0.064828	0.064828
PCB66	0.0073	2.00×10^{-10}	0.000010	1.76×10^{-8}	0.000882	1.78×10^{-8}	0.000892	0.000892
PCB99	0.026	7.12×10^{-10}	0.000036	6.28×10^{-8}	0.003141	6.35×10^{-8}	0.003177	0.003177
PCB87	0.042	1.15×10^{-9}	0.000058	1.01×10^{-7}	0.005069	1.02×10^{-7}	0.005127	0.005127
PCB109	0.0333	9.12×10^{-10}	0.000046	8.04×10^{-8}	0.004021	8.13×10^{-8}	0.004067	0.004067
PCB151	0.0113	3.10×10^{-10}	0.000015	2.73×10^{-8}	0.001367	2.76×10^{-8}	0.001383	0.001383
PCB153	0.022	6.03×10^{-10}	0.000030	5.32×10^{-8}	0.002659	5.38×10^{-8}	0.002689	0.002689
PCB146	0.0247	6.77×10^{-10}	0.000034	5.97×10^{-8}	0.002983	6.03×10^{-8}	0.003017	0.003017
PCB137	0.0207	5.67×10^{-10}	0.000028	5.00×10^{-8}	0.002501	5.06×10^{-8}	0.002529	0.002529
PCB187	0.0325	8.90×10^{-10}	0.000045	7.85×10^{-8}	0.003924	7.94×10^{-8}	0.003969	0.003969
PCB183	0.0233	6.38×10^{-10}	0.000032	5.63×10^{-8}	0.002813	5.69×10^{-8}	0.002845	0.002845
PCB180	0.03	8.22×10^{-10}	0.000041	7.25×10^{-8}	0.003625	7.33×10^{-8}	0.003666	0.003666
PCB170	0.0367	1.00×10^{-9}	0.000050	8.86×10^{-8}	0.004428	8.96×10^{-8}	0.004478	0.004478
PCB206	0.0567	1.55×10^{-9}	0.000078	1.37×10^{-7}	0.006836	1.38×10^{-7}	0.006914	0.006914

CDI, chronic daily intake; HQ, hazard quotient; HI, hazard index. HQ > 1 indicates potential non-carcinogenic risk. All calculated HQ values are < 1.0

Table 8 Evaluation of average daily exposure and hazard indices of PCBs for child exposure scenario (15 kg body weight)

Congener	ADD (mg/kg/day)	HQ
PCB1	1.28×10^{-9}	0.000064
PCB6	1.92×10^{-9}	0.000096
PCB18	6.30×10^{-8}	0.003149
PCB31	9.81×10^{-8}	0.004903
PCB44	3.39×10^{-8}	0.001696
PCB66	4.68×10^{-10}	0.000023
PCB99	1.66×10^{-9}	0.000083
PCB87	2.69×10^{-9}	0.000134
PCB109	2.13×10^{-9}	0.000107
PCB151	7.23×10^{-10}	0.000036
PCB153	1.41×10^{-9}	0.000070
PCB146	1.58×10^{-9}	0.000079
PCB137	1.32×10^{-9}	0.000066
PCB187	2.08×10^{-9}	0.000104
PCB183	1.49×10^{-9}	0.000075
PCB180	1.92×10^{-9}	0.000096
PCB170	2.35×10^{-9}	0.000117
PCB206	3.63×10^{-9}	0.000181

ADD, average daily dose; HQ, hazard quotient. HQ > 1 indicates potential non-carcinogenic risk for children. All calculated HQ values are < 1.0

Hazard Quotients exceeding 1.0 indicate potential for non-carcinogenic effects, but these estimates incorporate conservative exposure assumptions. In this corrected assessment, no HQ values exceeded 1.0. The elevated HQ values for PCB18, PCB31, and PCB44 relative to other congeners warrant targeted monitoring, though actual health outcomes depend on bioavailability, mixture effects, and individual susceptibility.

For the adult exposure scenario, ADD values vary between 5.48×10^{-10} and 2.70×10^{-8} mg/kg/day. PCB18, PCB31, and PCB44 have the greatest ADD values.

HQ values for children range from 0.000023 to 0.004903. The top three HQ values are for PCB31, PCB18, and PCB44. Notably, all HQ values remain below one, indicating that non-carcinogenic risks are within acceptable regulatory limits.

The detection of fungal genera in untreated groundwater warrants microbiological monitoring and further investigation of their potential public-health significance; this study did not test fungal viability, pathogenicity, toxin production or dose-response relationships, so established mycotoxicosis or opportunistic-infection risk cannot be concluded from these data (cf. Barakat et al. [7]; Ediagbonya [11]; Ugochukwu [35]). The detectable PCB congener pattern is consistent with anthropogenic inputs, though source attribution was not formally tested in this study. Every day, the public collects a substantial volume of water from the subsurface sources under investigation. According to Hopf et al. [16], the progressive deterioration of these contaminants may increase their ecological persistence, affecting human immunological, reproductive, neurological, and endocrine systems. Increased exposure to contaminants can have comparable effects on thyroid hormone levels in neonates and expecting mothers.

3.5 Integrated implications for water safety

To explore the association between biological and chemical contamination, we examined the spatial co-occurrence of fungal loads and PCB concentrations across the

sampling locations. Although a formal statistical correlation was constrained by the aggregated nature of the quantitative fungal data, a potential spatial co-occurrence is observed. Notably, the Igbokoda Health Centre (IGHC) emerged as a distinct hotspot, exhibiting both the highest total PCB concentrations and significant fungal prevalence. This spatial overlap is a hypothesis requiring verification rather than an established finding: no hydrochemical tracing, source-apportionment modelling, well-construction analysis, microbial source tracking or contaminant-transport modelling was undertaken, so a common source (e.g., proximity to unlined landfills, e-waste dumping, or compromised well casings) cannot be confirmed from spatial co-occurrence alone. If confirmed by future targeted studies, concurrent biological and chemical stressors at hotspot locations could pose complex health risks, particularly for immunocompromised individuals, underscoring the need for integrated water quality management and further investigation.

While measured PCB concentrations and modelled non-carcinogenic health risks were found to be within acceptable regulatory limits, the concurrent occurrence of fungal genera exceeding commonly cited but non-regulatory reference values warrants further microbiological monitoring and investigation. The fungal isolates, particularly genera such as *Aspergillus*, *Fusarium*, and *Penicillium*, that are capable of mycotoxin production in some contexts, were detected, though pathogenicity, viability and toxin production were not directly tested in this study and mycotoxicosis/opportunistic-infection risk is not demonstrated. Meanwhile, PCB contamination presents non-carcinogenic risk through the ingestion and dermal pathways evaluated here; carcinogenic risk was not assessed.

It is important to keep three distinct concepts separate when interpreting these results. First, measured PCB concentrations remained below the U.S. EPA regulatory benchmark of 500 ng/L. Second, modelled non-carcinogenic hazard quotients for individual congeners were below 1.0, which represents a separate risk metric with its own exposure assumptions. Third, microbiological water quality was assessed qualitatively and is not established by either of the PCB-based metrics. None of these three findings alone establishes that the water is generally safe or unsafe. Although the U.S. EPA maximum allowable quantity of 500 ng/L for PCBs in drinking water was not exceeded, the presence of any detectable PCBs implies anthropogenic groundwater contamination that warrants continuous monitoring. Cumulative non-carcinogenic risk, as represented by the Hazard Index in Supplementary Table S4, was consistently higher for children than adults across all sampling locations due to greater ingestion rates per unit body weight. This cumulative metric provides a more appropriate basis for characterizing overall population risk than single-congener maxima. The spatial heterogeneity in contamination patterns, with hotspots such as IGHG showing elevated levels of both fungal and PCB contaminants, suggests localized pollution sources that require source identification and remediation. The dominance of lower-chlorinated PCB congeners in groundwater is consistent with preferential mobility of these compounds through sandy aquifer materials, though this pattern does not independently establish recent anthropogenic inputs.

The spatial heterogeneity in contamination patterns, with hotspots such as IGHG showing elevated levels of both fungal and PCB contaminants, suggests localized pollution sources that require source identification and remediation. The dominance of

lower-chlorinated PCB congeners in groundwater indicates either recent contamination events or preferential mobility of these compounds through sandy aquifer materials.

Immediate interventions should include: (1) provision of alternative safe groundwater for affected communities, particularly for vulnerable populations including children and pregnant women; (2) implementation of point-of-use water treatment systems capable of removing both microbial and chemical contaminants; (3) regular monitoring of groundwater quality with expanded analyte lists including mycotoxins and PCB metabolites; and (4) public health education on risks associated with untreated groundwater consumption.

Long-term strategies must address source control through proper management of PCB-containing equipment, improvement of waste disposal infrastructure, and protection of groundwater recharge areas from contamination. The findings underscore the need for integrated water quality management that considers both biological and chemical contaminants in assessing water safety for human consumption.

4 Conclusion and recommendations

4.1 Conclusion

This study provides an integrated occurrence assessment of fungal genera and PCBs in coastal groundwater of Ondo State, Nigeria. The findings reveal that measured PCB concentrations and associated non-carcinogenic health risks remain within acceptable regulatory limits, while the detection of fungal genera — including *Aspergillus*, *Fusarium*, and *Penicillium* — across sampling locations warrants microbiological monitoring and further investigation of their potential public-health significance; pathogenicity and toxin production were not demonstrated in this study. It is important to note that species-level identifications presented are tentative, based on morphological characteristics alone. The dominance of lower-chlorinated PCB congeners is consistent with, but does not on its own establish, recent anthropogenic inputs such as localized e-waste or transformer oil disposal; source apportionment was not performed. A key limitation of this study is that sampling was conducted exclusively during the dry season; therefore, the findings may not fully represent annual exposure dynamics, as rainfall during the wet season could alter contaminant transport through dilution or increased runoff. Furthermore, carcinogenic risk could not be evaluated because the necessary polychlorinated biphenyl congeners were not measured, limiting the assessment to non-carcinogenic effects. Future investigations should include carcinogenic PCB congeners to provide a more comprehensive assessment of potential human health risks. Future research should incorporate seasonal monitoring, molecular fungal identification, mycotoxin and pathogenicity testing, and bioavailability-adjusted risk assessments. As a precautionary measure pending confirmatory microbiological and chemical assessment, point-of-use water treatment and continued groundwater quality monitoring are recommended to safeguard public health.

4.2 Recommendations

Based on these findings we recommend: (1) implementation of routine monitoring programs for both microbial and chemical contaminants in coastal groundwater; (2) point-of-use treatment interventions such as chlorination and activated carbon filtration for households relying on contaminated sources as general public health measures;

(3) source protection measures to reduce PCB inputs from informal waste disposal and industrial activities; and (4) community health education on safe water handling practices. Future research should incorporate molecular fungal identification and bioavailability-adjusted risk assessment to refine exposure estimates.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43832-026-00460-w>.

Supplementary Material 1

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Author contributions

All authors contributed to the study conception and design. Ikuesan Felix Adeleke collected the samples did some of the analyses, discussion. Ediagbonya Thompson Faraday did some of the discussion and proofread the work. The first draft of the manuscript was written by Ikuesan Felix Adeleke and all authors proofread the manuscript and approved the final manuscript for publication.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants or animal subjects.

Consent to publish

Not applicable.

Competing interests

The authors declare no competing interests.

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